

DeepSeek-V2: A Strong, Economical, and Efficient MoE Language Model

236B total parameters · 21B activated per token · 128K context length

arXiv:2405.04434 · DeepSeek-AI

42.5%

Training cost reduction vs. DeepSeek 67B

93.3%

KV cache reduction via MLA

5.76×

Generation throughput improvement

The LLM Scaling Dilemma: Performance vs. Training Cost and Inference Speed

Larger models yield better intelligence, but at prohibitive cost

Training Cost

Dense parameter scaling makes pre-training budgets explode. A 70B-class dense model requires massive GPU clusters and weeks of training time.

KV Cache Bottleneck

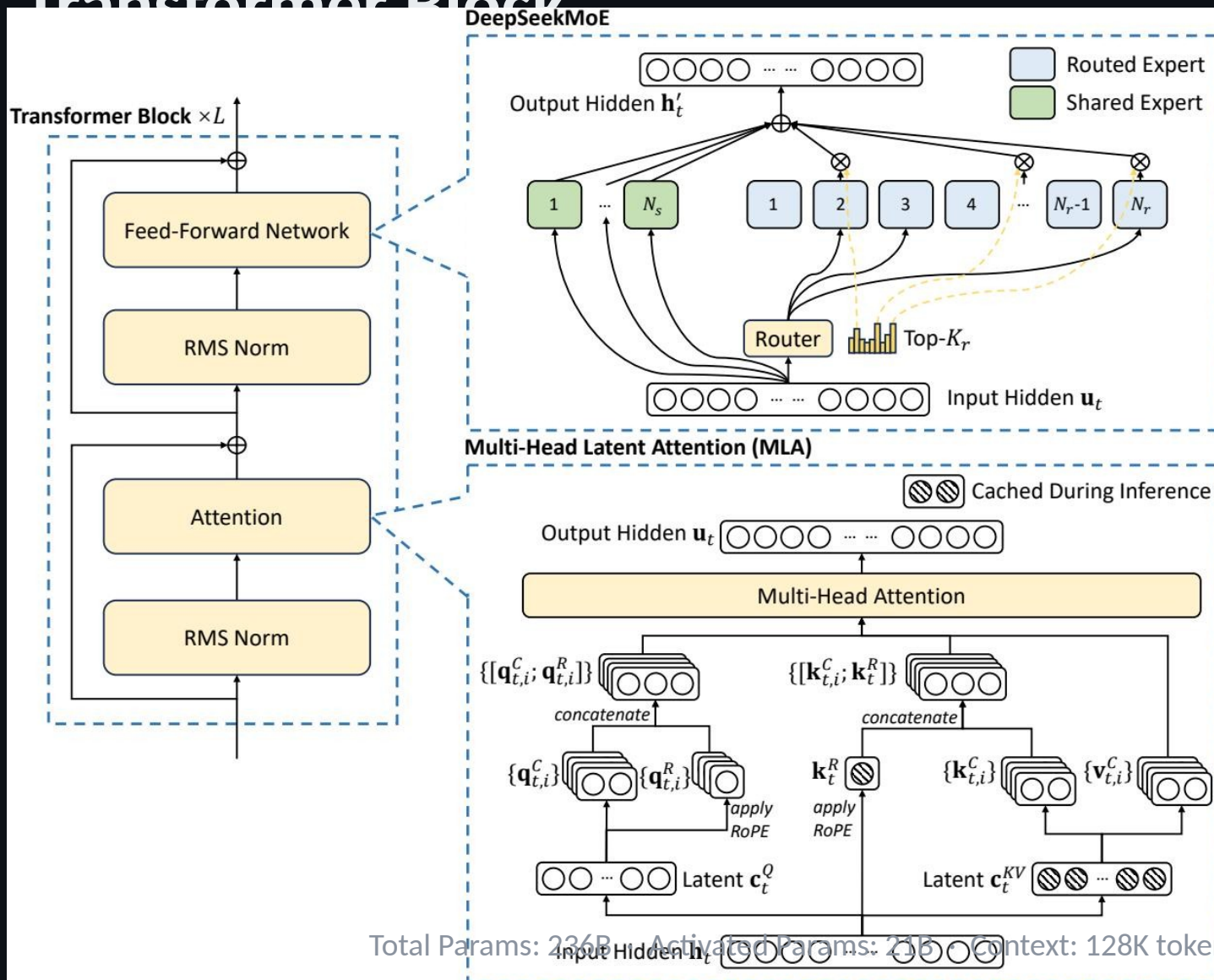
Standard Multi-Head Attention stores full key-value pairs per head per token, creating memory pressure that cripples long-context inference.

Inference Throughput

Memory-bound generation limits serving capacity. Every millisecond of latency translates directly to hardware cost at scale.

Core Insight → MoE sparsity + compressed attention can break the Pareto frontier: achieve higher quality at lower training and inference cost.

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in a Transformer Block



Transformer Block Components

► Multi-Head Latent Attention (MLA)

Replaces MHA with low-rank KV compression. Caches a compact latent vector instead of full K/V per head.

► DeepSeekMoE FFN

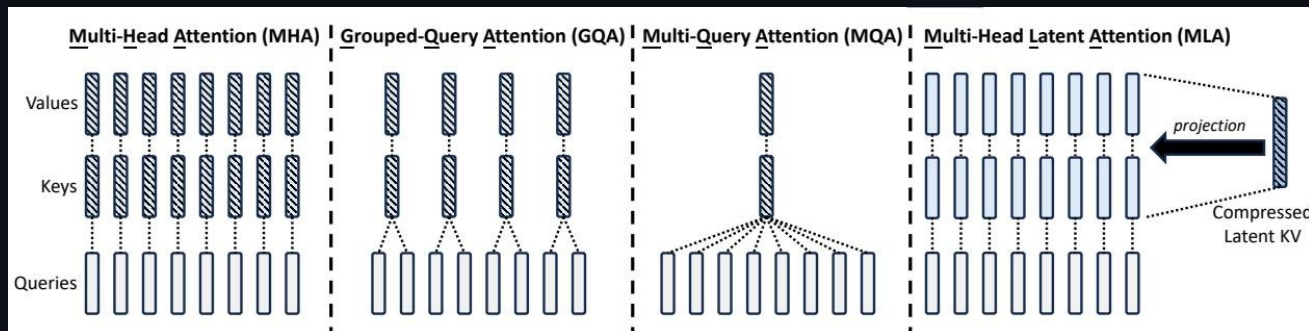
Shared experts + routed experts. Only a subset of 236B params are activated per token (21B).

► Decoupled RoPE

Positional information injected via a separate key projection, preserving compression efficiency.

Multi-Head Latent Attention Compresses KV Cache by 93.3%

Better performance than MHA with dramatically less cache — strictly dominating GQA and MQA



How MLA Works

- Low-rank key-value joint compression: instead of caching full K and V vectors per head, MLA compresses them into a compact latent vector c_t^{KV} .
- During inference, only the latent vector is cached — full keys and values are reconstructed on-the-fly via learned projection matrices.
- A decoupled RoPE-based positional key is maintained separately, preserving positional information without inflating cache size.
- Result: 93.3% KV cache reduction vs. standard MHA, with superior downstream performance.

MHA → high cache, best perf | GQA/MQA → lower cache, degraded perf | MLA → minimal cache, best perf

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Two expert types, device-limited routing, and auxiliary losses for load-balanced, communication-efficient training

Shared Experts (N_s)

Always activated for every token. Capture common, high-frequency knowledge across the corpus, reducing redundant routing overhead.

Routed Experts (N_r)

Sparsely activated via a top-Kr router. Fine-grained segmentation increases specialization potential — each expert masters a narrow domain.

Device-Limited Routing

Each token is constrained to experts on at most M devices. This bounds all-to-all communication overhead across GPU nodes.

Load-Balancing Losses

Auxiliary losses enforce balance across experts and devices. A token-dropping strategy during training prevents expert overflow without affecting inference.

Result: 236B total parameters with only 21B activated per token — 11.2× parameter scaling at ~1/3 the active compute of a dense 70B model.

42.5% Training Cost Reduction and 5.76× Generation

Throughput vs. DeepSeek 67B

Relative efficiency improvements of DeepSeek V2 over its dense predecessor DeepSeek 67B

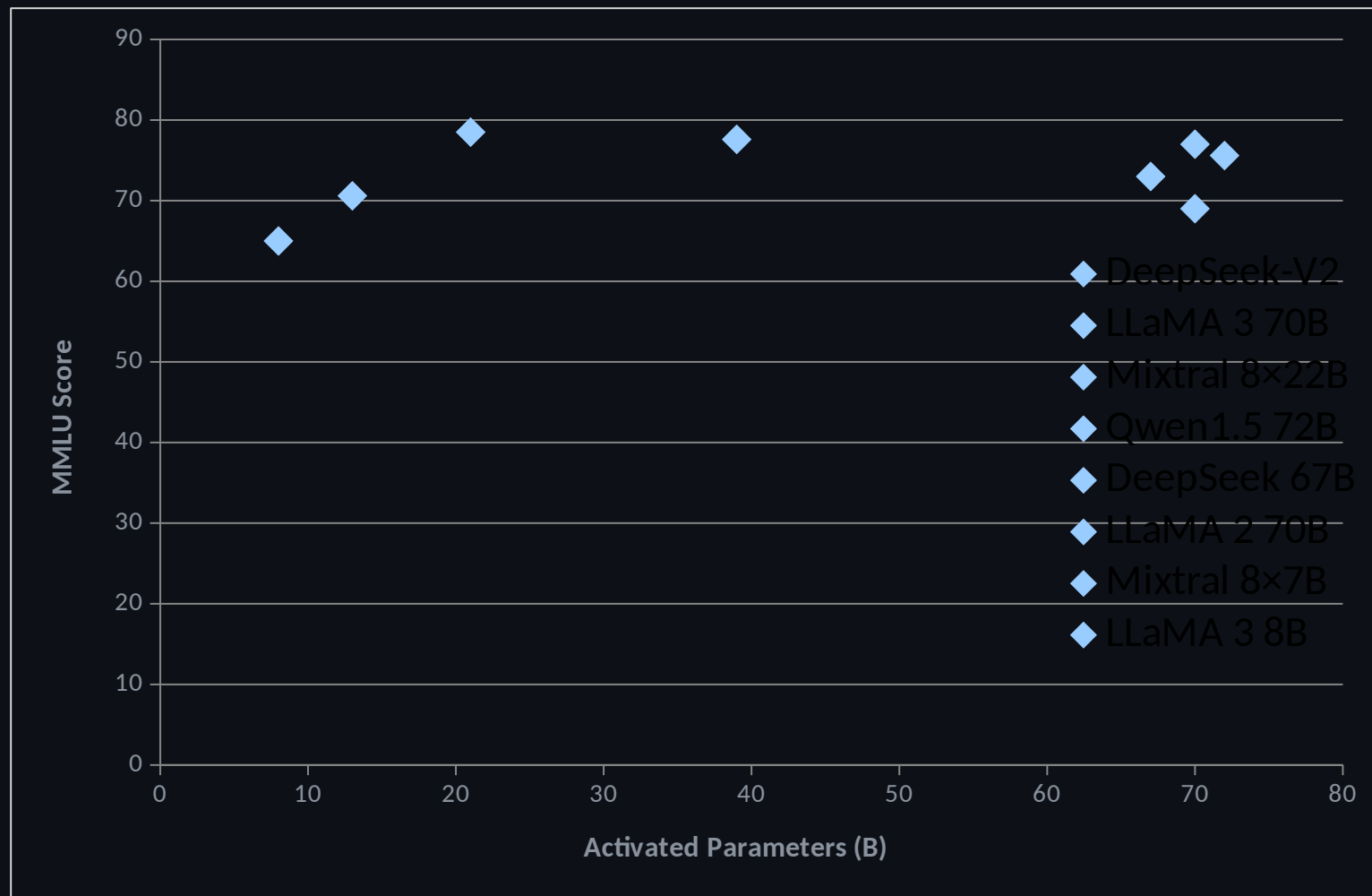


All metrics are relative improvements reported in the DeepSeek-V2 paper. No absolute FLOPs, cache sizes, or throughput numbers are stated.

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B

Activated Parameters

Pareto optimal positioning: highest MMLU score with the fewest activated parameters



Why This Matters

- DeepSeek-V2 achieves the highest MMLU score (78.5) in this comparison.
- It does so with only 21B activated parameters — 3.3× fewer than LLaMA 3 70B and 1.9× fewer than Mixtral 8x22B.
- This is a clear Pareto improvement: better quality at lower active compute.
- Dense models like LLaMA 2 70B and DeepSeek 67B are dominated on both axes.

Data sourced from the DeepSeek-V2 paper (arXiv:2405.04434). MMLU scores are 5-shot averages where applicable.

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

Post-training: SFT on 4.5M conversational sessions + GRPO-based Reinforcement Learning

AlpacaEval 2.0

38.9%

Length-controlled win rate

Competitive in English open-ended evaluation against strong open-source chat models.

MT-Bench

8.97

Overall score

Strong generalist conversation quality across diverse instruction-following tasks.

AlignBench (Chinese)

7.91

Overall score

Outperforms all open-source models and most closed-source models in Chinese alignment quality.

Key Result → DeepSeek-V2 Chat (RL) is the strongest open-source chat model in Chinese and competitive with closed-source systems in English. The alignment pipeline (SFT + GRPO) demonstrates that architectural efficiency gains translate directly into high-quality conversational models.

Key Takeaways: Architectural Innovation Unlocks Efficient

DeepSeek-V2 demonstrates that the performance–cost Pareto frontier is not fixed

Scaling

- 1 MLA is a strictly better attention mechanism**
Achieves superior performance to standard MHA while reducing KV cache by 93.3%. It dominates GQA and MQA on both quality and efficiency.
- 2 DeepSeekMoE enables economical training at massive scale**
A 236B-parameter MoE trains at 42.5% lower cost than a 67B dense model, thanks to fine-grained experts, shared expert isolation, and efficient routing.
- 3 Top-tier performance with minimal activated compute**
Only 21B activated parameters match or exceed 70B-class dense models on MMLU — a clear Pareto-optimal result.
- 4 Strong multilingual alignment**
DeepSeek-V2 Chat (RL) leads open-source models in Chinese (AlignBench 7.91) and is competitive in English open-ended evaluation.
- 5 Practical deployment advantages**
5.76× generation throughput improvement and 93.3% KV cache reduction make DeepSeek-V2 practical for large-scale production serving.